

(a) Baseline A (R0)

open-weight

closed-source API

Sequential workflow	39	37	37	5	1	9	4	1	1	18	47	33	12
Power flow	24	24	31	18	19	9	9	1	1	25	44	26	8
Optimal power flow	41	30	38	5	33	0	5	6	0	21	38	22	5
DC power flow	12	26	4	2	5	4	0	0	0	12	58	12	1
Diagnose and repair	24	18	23	12	5	16	2	0	3	13	29	27	18
Time-series control	11	28	21	9	1	0	0	0	0	23	24	9	2
OPF comparison	23	16	1	2	5	0	0	0	0	22	29	6	0
Model comparison	22	25	23	5	11	10	2	1	1	10	29	18	10
Contingency analysis	0	10	25	0	0	1	1	0	0	1	8	5	0
Parallel sweep	0	0	0	0	0	0	0	0	0	0	5	0	0
Three-phase short circuit	1	0	0	0	0	2	0	0	0	0	7	0	0
State estimation	0	4	0	1	0	0	0	0	0	9	30	0	0
Power flow then SC	0	0	0	0	0	0	0	0	0	0	4	0	0
Two-phase short circuit	0	0	0	0	0	0	0	0	0	0	2	0	0
Contingency correction	3	3	6	0	1	0	0	0	0	4	2	1	1

Llama-405B
Qwen3-480B
OSS-120B
Qwen3-Next
Llama-70B
Qwen2.5-32B
Qwen2.5-14B
Llama-8B
Qwen2.5-7B
Claude
GPT-5.4
DeepSeek
Gemini

(b) Full method C+FDRS (R3)

open-weight

closed-source API

Sequential workflow	87	85	87	86	86	82	84	71	73	85	85	85	85
Power flow	76	77	79	74	73	74	74	54	59	76	77	71	76
Optimal power flow	82	79	73	72	77	65	64	52	50	78	59	64	73
DC power flow	82	80	82	81	78	74	80	35	45	82	80	72	81
Diagnose and repair	48	45	61	45	47	56	47	63	35	53	76	49	65
Time-series control	77	67	62	70	56	65	52	6	16	61	56	60	71
OPF comparison	64	65	51	53	58	47	52	33	37	64	66	60	60
Model comparison	49	60	48	51	53	52	47	40	26	54	53	46	54
Contingency analysis	60	62	63	53	61	26	22	7	39	72	80	67	48
Parallel sweep	76	72	65	74	71	70	30	38	8	50	40	53	19
Three-phase short circuit	63	59	46	56	52	45	55	40	21	49	53	46	32
State estimation	70	73	68	24	24	27	26	7	39	75	57	47	40
Power flow then SC	61	59	43	57	55	41	32	39	10	59	59	39	27
Two-phase short circuit	71	61	39	60	52	37	44	35	24	53	41	38	23
Contingency correction	63	37	28	27	9	10	9	3	6	46	47	40	24

Llama-405B
Qwen3-480B
OSS-120B
Qwen3-Next
Llama-70B
Qwen2.5-32B
Qwen2.5-14B
Llama-8B
Qwen2.5-7B
Claude
GPT-5.4
DeepSeek
Gemini

